Curriculum Vitae

Computer scientist working on machine learning for electronic health records and on EEG-based brain–computer interfaces. Currently applying for MPhil and PhD positions starting in 2027.

Education

Shandong Normal University — B.Eng. in Computer Science and Technology Jinan, China

- GPA 3.94 / 5.0 (89.4 / 100), top 10%; First-class Scholarship and Outstanding Student.
- Thesis: *Dynamic Graph and Multi-view Drug Information for Safe Medication Recommendation* (grade 95), supervised by Prof. Xiaomei Yu.

Research Interests

- EEG-based brain–computer interfaces: cross-subject decoding, self-supervised brain foundation models, interpretable and closed-loop BCI.
- Large language models: evidence-aware generation, misinformation and rumor detection.
- Machine learning for electronic health records: dynamic graphs, multi-view representation learning, safe medication recommendation.

Publications

> **ENCMed: A 3D-Encoder Framework with Neighbor Contrastive Learning for Safe Medication Recommendation**
> first author
> **G. Liu**, Z. Liu, X. Yu, X. Li, X. Zheng, X. Fan
> *Manuscript under review*, 2026.
> A three-branch encoder that fuses molecular fingerprints with drug–drug interaction graphs and a patient's visit sequence, trained with neighbour contrastive learning so that the recommended combination is both accurate and safe.

> **CLEAR: Prototype-conditioned flow purification for LLM-based rumor detection with Dirichlet evidential learning**
> Z. Liu, H. Wu, X. Fang, **G. Liu**, et al.
> *Information Processing & Management* IF 8.1, 2026. [DOI]
> Conditions a flow-matching generative model on prototype representations and adds Dirichlet evidential learning, so that an LLM-based detector returns calibrated uncertainty instead of a bare label.

> **DNMDR: Dynamic Networks and Multi-view Drug Representations for Safe Medication Recommendation**
> first author
> **G. Liu**, Z. Liu, X. Yu, X. Li, X. Zheng, X. Fan
> *Knowledge-Based Systems* IF 8.0, 2025. [DOI]
> Rebuilds a patient's history as a weighted dynamic network whose edges are inferred from conditional probabilities, and represents drugs from several views at once so that drug–drug interactions are penalised during combination prediction.

Transformer-based medication recommendation with a multiple graph augmentation strategy co-author

X. Li, X. Yu, **G. Liu**, X. Fan, F. Ge, Y. Zhao, X. Zheng

Expert Systems with Applications IF 8.2, 2024. [DOI]

Applies a Transformer over augmented drug and diagnosis graphs to capture higher-order relations between medical codes before predicting the medication combination.

DG-MRM: Dynamic Graph and Multi-view Medication Information for Recommending Medication Combination first author

G. Liu, Z. Liu, X. Yu, X. Li, X. Zheng, X. Fan

KDD-AIDSH 2024 — AI and Data Science for Healthcare workshop, 2024. [OpenReview]

The workshop version of the dynamic-graph medication recommendation line, reporting the first results on MIMIC-III.

Research Grants

- **Principal investigator**, *Research on medication recommendation method based on multi-view dynamic graph contrastive learning*, National Undergraduate Training Program for Innovation and Entrepreneurship, 05/2023 – 05/2025.
- **Principal investigator**, *Research on interpretable deep learning approach for electronic health records*, Shandong Normal University Training Program for Innovation and Entrepreneurship, 05/2022 – 05/2023.
- **Principal investigator**, *Research on human–computer interaction methods for NAO robots*, Shandong Normal University Undergraduate Research Fund, 11/2021 – 03/2023.

Research Experience

Explainable deep learning methods for electronic health records 05/2023 – 08/2026

Prof. Xiaomei Yu, Shandong Normal University

Used large language models to analyse medical data in EHR, and built accurate drug recommendation methods for electronic health records using temporal graph structures. Outputs include a medication recommendation software copyright V1.0 registered on 2024-03-29.

Fake news detection based on large language models 02/2023 – 05/2026

Assoc. Prof. Hongchen Wu, Shandong Normal University

Analysed semantic interactions between posts and comments in social-media datasets to explore how ambiguous event descriptions influence emotional responses to real and fake news; proposed innovative solutions and implemented the algorithms. Outputs include a National Second Prize at the 16th China University Computer Design Competition.

Intelligent classroom management system based on edge computing 05/2022 – 05/2023

Dr. Ke Liu, Shandong Normal University

Developed a sound-source localisation and tracking device using a Raspberry Pi 4B and infrared laser technology, and designed a contactless temperature measurement, mask detection and facial recognition system. Outputs: Second Prize, 2022 Shandong Province College Student Electronic Design Competition; First Prize, Shandong Normal University Electronic Design Competition; patent granted as first inventor (2022-10-19, No. 202222749613.8).

Research on human–robot interaction methods for NAO robots 07/2021 – 06/2022

Assoc. Prof. Wei Yan, Shandong Normal University

Developed an intelligent obstacle avoidance system for NAO robots using computer vision techniques, and designed a music-domain human–robot interaction system using natural language processing and knowledge-graph methods. Outputs: Third Prize, 24th China Robotics and Artificial Intelligence Competition (Shandong); national computer software copyright for the intelligent robot obstacle avoidance system V1.0; patent granted as first inventor (2023-10-24, No. ZL 2023 2 0465706.X).

Professional Experience

Emergency Management Bureau, Tengzhou — Intern

- Dispatched information from the bureau to relevant township and subdistrict officials.
- Assisted in designing and coding a small system for automating duty scheduling.

Shandong Province Artificial Intelligence Education Base — Course Developer

- Guided visiting students on research tours, explaining 3D printing, computer-vision based waste sorting and colour-block sorting devices, and a computer-vision powered Tic-Tac-Toe robot.
- Assisted in the design of a computer-vision based waste management system and colour sorting devices using the TensorFlow framework.

Skills

- **Programming:** Python, C, C++, Java, MATLAB, LINGO.
- **Frameworks:** PyTorch, TensorFlow.
- **Other:** web development; certified in the Industrial and Information Technology Talent Professional Knowledge Assessment — Mathematical Modeling.
- **Languages:** Mandarin Chinese (native), English (fluent).

Awards

- Second Prize, 16th China University Computer Design Competition (national), 2023.
- Second Prize, National College Student Mathematical Modeling Competition (national), 2022.
- Third Prize, 12th MathorCup Mathematical Modeling Challenge — Big Data (national), 2022.
- Third Prize, 12th MathorCup Mathematical Modeling Challenge (national), 2022.
- Third Prize, 3rd Huashu Cup National College Student Mathematical Modeling Competition (national), 2022.
- Second Prize, 13th Blue Bridge Cup (Shandong), 2022.
- Second Prize, National College Student Electronic Design Competition (Shandong), 2022.
- Third Prize, 24th China Robotics and Artificial Intelligence Competition (Shandong), 2022.
- Third Prize, China Collegiate Computing Contest — Network Technology Challenge (East China), 2023.

- First-class Scholarship and Outstanding Student, 2021; Second- and Third-class Scholarships, Outstanding Student, 2022–2023; Outstanding Communist Youth League Member, 2023.